Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/12/2022
PI Name:	Porada, Christopher Ph.D.		
Project Title:	Using Human Organoids and Fossilized Remains from Extinct Hominins to Unlock the Secrets of Torpor/Hibernation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27157	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 BRASH2101: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	07/01/2022	**End Date:**	03/31/2026
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No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Performance Goal No.:	
Performance Goal Text:	
Task Description:	Deep space missions, such as those NASA has planned to Mars, will subject astronauts to extreme physiological and psychological stressors. Such missions will also require multi-year onboard life support for the crew. While advanced strategies for onboard generation of required metabolic inputs and recycling waste may one day reduce this burden on storage capacity, such interventions will not address the physiological and psychological stressors of such missions. The ability to induce astronauts into a hypometabolic state would reduce the psychological effects of isolation and containment, diminish the deleterious physiological effects of microgravity and radiation, and dramatically reduce O2, H2O, and food consumption and the consequent production of metabolic wastes. Studies performed in a variety of animal species that hibernate or enter daily torpor provide invaluable insight into the involved molecular and biochemical processes, but also underscore the marked species differences in pathways used by natural hibernators. Also, compounds identified to successfully induce torpor in non-hibernators (e.g., Hydrogen sulfide/H2S) are often toxic to humans. In the pharmaceutical industry, preclinical animal tests often do not accurately predict efficacy or identify toxicities. These limitations led Defense Threat Reduction Agency (DTRA) to create the X.C.E.L. program to develop clinically relevant microfluidic-based systems of integrated human organoids to be used as a more reliable surrogate to human pathophysiology than animals. The Wake Forest Institute for Regenerative Medicine (WFIRM)-led X.C.E.L. program resulted in a human “body-on-a-chip” in which microengineered organ equivalents (MOEs) of all major tissues, created with primary human cells, can be maintained long-term and retain normal functionality, and have accurately identified toxicities of former Food and Drug Administration (FDA)-approved drugs later recalled due to serious adverse events (SAEs). MOEs include a normal repertoire of human cell types, tissue-specific extracellular matrix (ECM), and native stiffness, enabling them to accurately mimic in vivo cell-cell, cell-ECM, and mechanical interactions. MOEs have also successfully modeled the metabolic state of healthy and diseased human tissue. They thus provide an ideal platform to accomplish our overall goals, which are to: 1) identify novel methods for inducing a hypometabolic state in human tissues; 2) define the optimal parameters for achieving such a state while preserving human tissue viability and function; 3) model cellular adaptation to reduced temperature, nutrient, and energy inputs; 4) identify molecular pathways associated with these adaptations; 5) determine means to safely reduce the metabolic needs of 3 key human tissues (heart, liver, and bone marrow) for prolonged periods and subsequently restore them to a normal metabolic state; and 6) test the ability of a hypometabolic state to protect these human tissues from the deleterious effects of space radiation. We will also exploit the recent finding that an extinct species of humans (SH hominins) had the ability to hibernate by performing genomic analyses comparing this species to closely related non-hibernators (Neanderthals and Denisovans) to unlock clues to human hibernation and use the obtained information to further optimize induction of a hypometabolic state in the human MOEs. This proposal is directly responsive to the key objectives and research emphases of BRASH2101 and addresses multiple risks identified in the NASA Human Research Program (HRP) Roadmap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2022.
Bibliography Type:	Description: (Last Updated: 08/05/2025)